

AI-Powered Investment Intelligence Using NIFTY-50 Market Data

Objective

This submission builds a local investment decision-support platform using only the provided NIFTY-50 stock archive, stock metadata, and NSE index datasets. The emphasis is portfolio construction, risk assessment, explainability, anomaly monitoring, and actionable investment support rather than standalone price prediction.

Dataset

The pipeline processed 235,192 equity rows across 49 current NIFTY-50 stocks from 2000-01-03 to 2021-04-30. Metadata contributes company and industry labels. NIFTY 50 and INDIA VIX files from the supplied second archive provide benchmark return, beta, and market-regime context.

Architecture

A reproducible Python pipeline loads CSVs, engineers features, trains time-split models, builds portfolio profiles, detects anomalies, and writes deterministic artifacts. A Streamlit application reads those artifacts for interactive stock intelligence, portfolio construction, risk analysis, and explanation views.

Constraint Compliance

No live market data, financial APIs, news, sentiment, proprietary feeds, or alternative external datasets are used. All calculations are derived from local files supplied in the project directory.

EDA and Feature Engineering

Exploratory Findings

The stock universe spans multiple industries including financial services, energy, automobiles, consumer goods, IT, metals, pharma, cement, telecom, services, construction, fertilizers, and media. The long history supports cycle-aware risk estimation and post-2020 out-of-sample validation.

Features

Daily close, high, low, VWAP, volume, turnover, and benchmark series are transformed into return horizons, annualized volatility, moving-average gaps, EMA spread, MACD, RSI, Bollinger z-score, ATR, volume and turnover z-scores, drawdown, rolling beta to NIFTY 50, and VIX regime level.

Targets

The predictive target is 21-trading-day forward return. A second binary target captures whether the 21-day forward return is positive, supporting directional decision-making.

Validation Design

Training uses observations before 2020-01-01. Validation uses 2020 onward, preserving temporal order and avoiding look-ahead leakage.

Modeling Methodology and Explainability

Predictor Engine

A HistGradientBoostingRegressor forecasts 21-day return. A RandomForestClassifier estimates probability of positive 21-day return. Both use engineered technical, risk, liquidity, benchmark, and regime features.

Evaluation

Out-of-sample MAE is 10.18%, RMSE is 14.45%, R2 is -0.003, regression directional accuracy is 55.72%, and classifier directional accuracy is 55.56%.

Explainability

The platform exposes global permutation importance, latest stock-level feature values, transparent stock rationales, risk metrics, anomaly tags, and portfolio weight evidence. This creates auditable recommendations rather than black-box buy lists.

Limitations

The models learn from historical technical and index-derived behavior only. They do not include earnings, macro releases, news, analyst revisions, transaction costs, taxes, or live liquidity changes. Recommendations are decision-support outputs, not financial advice.

Top Ranked Stock Opportunities

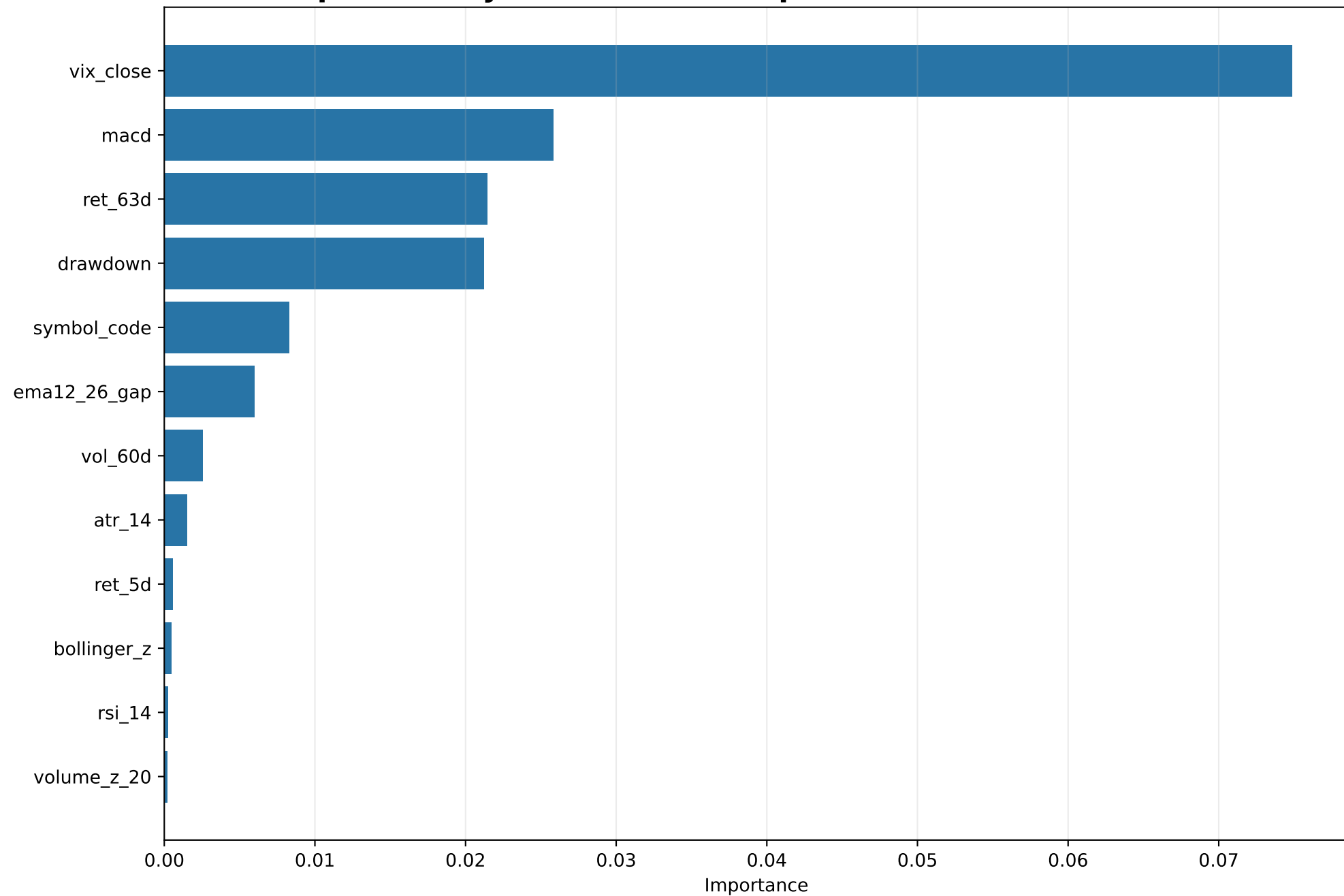
Ranking combines 21-day forecast, probability of upside, recent Sharpe ratio, and volatility penalty.

source_file_symbol	industry	quality_score	forecast_return_21d	forecast_up_probabil	sharpe	volatility	max_drawdown
GRASIM CEMENT	CEMENT & CEMENT PROD	83.37%	4.63%	52.10%	5.48	33.18%	-12.13%
JSWSTEEL	METALS	83.37%	5.63%	52.33%	8.67	40.40%	-11.88%
VEDL	METALS	80.51%	5.64%	54.43%	4.81	48.79%	-34.26%
HINDALCO	METALS	78.78%	6.16%	52.06%	5.04	46.69%	-20.05%
BAJAJ-AUTO	AUTOMOBILE	78.47%	2.20%	55.54%	1.83	30.39%	-17.37%
ADANI PORTS	SERVICES	76.43%	2.42%	54.19%	4.19	39.96%	-14.28%
BAJFINANCE	FINANCIAL SERVICES	74.59%	6.41%	51.95%	3.06	47.30%	-22.50%
SBIN	FINANCIAL SERVICES	67.35%	3.54%	51.13%	2.22	41.41%	-21.57%
ULTRACEMENT	CEMENT & CEMENT PROD	66.94%	1.40%	52.91%	2.72	31.29%	-12.96%
TATAMOTORS	AUTOMOBILE	66.43%	6.75%	48.90%	4.99	56.52%	-19.13%
SHREECEMENT	CEMENT & CEMENT PROD	65.31%	1.89%	53.93%	1.43	33.47%	-20.64%
INDUSINDBK	FINANCIAL SERVICES	61.73%	4.00%	50.75%	1.82	54.70%	-28.18%

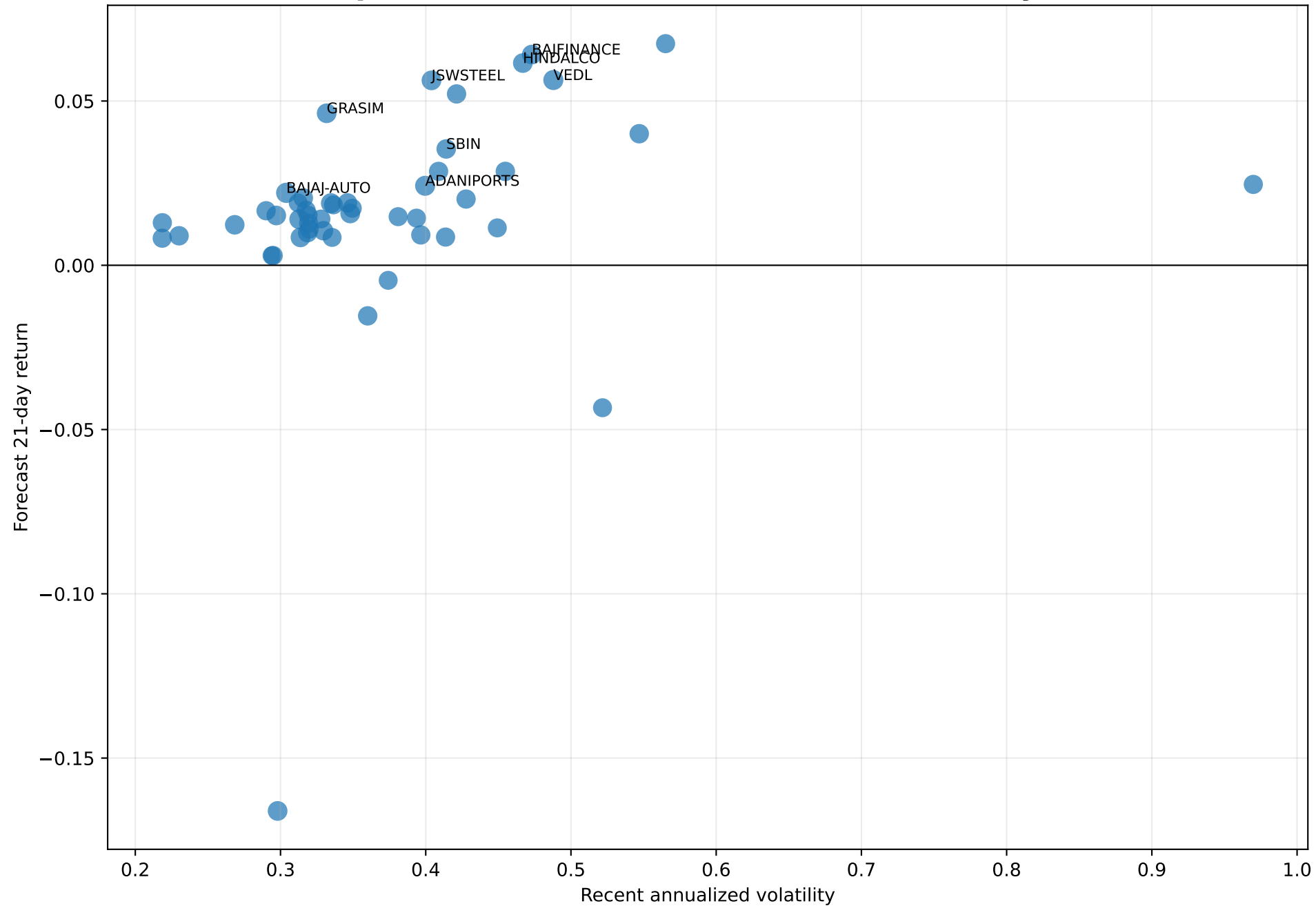
Portfolio Profiles and Risk Metrics

profile	expected_21d_return	expected_annualized_return	volatility	sharpe	sortino	max_drawdown	holdings
Conservative	3.87%	57.69%	27.37%	0.72	0.85	-51.31%	18
Balanced	2.00%	26.82%	20.85%	0.66	0.76	-36.47%	18
Aggressive	4.21%	64.04%	30.90%	0.76	0.93	-54.73%	18

Global Explainability: Permutation Importance



Investment Map: Forecast Return vs Historical Volatility



Portfolio Construction, Risk, and Insights

Construction Logic

The system creates Conservative, Balanced, and Aggressive long-only portfolios. Each profile applies different volatility tolerance, maximum position size, and risk-aversion penalty. Candidate stocks must pass model-quality filters and are weighted by expected return, probability of upside, Sharpe ratio, and volatility risk.

Risk Assessment

Every stock and portfolio is evaluated using annualized return, volatility, Sharpe ratio, Sortino ratio, maximum drawdown, Calmar ratio, and beta where relevant. Portfolio risk is calculated from the historical weighted return stream of selected holdings.

Anomaly Detection

The anomaly module flags unusual volume, volatility spikes, and extreme drawdown states using rolling z-scores and historical drawdown thresholds. These signals help users separate ordinary ranking changes from unusual market behavior.

Key Insight

The most useful decision signal is not a single forecast. The platform combines forecast direction, return magnitude, risk-adjusted performance, volatility, drawdown, and diversification to produce actionable but explainable investment choices.